

Notes: Collard-Wexler (Econometrica, 2013)

Demand Fluctuations in the Ready-Mix Concrete Industry

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1 Big picture

- **Question.** How do local demand fluctuations affect entry, exit, investment, market structure, and welfare in a geographically segmented oligopoly?
- **Industry setting.** Ready-mix concrete is a local, homogeneous, perishable product. Trucks cannot travel far before concrete becomes unusable, so counties approximate independent local oligopoly markets.
- **Demand source.** Concrete demand is driven mainly by local construction employment and public construction activity. Because construction activity fluctuates strongly across counties and time, the industry is a good setting for studying demand volatility.
- **Model.** The paper estimates a dynamic oligopoly model of entry, exit, and discrete plant-size investment. Plants can be out, small, medium, or large, and firms face fixed costs, competition effects, plant-size transition costs, and idiosyncratic shocks.
- **Estimation strategy.** The model is estimated with an indirect inference conditional-choice-probability algorithm. The approach uses observed entry, exit, plant-size transitions, and market structure to recover structural primitives.
- **Counterfactual policy.** The main counterfactual smooths short-run local construction demand over five-year windows while preserving long-run demand movements.
- **Main dynamic result.** Demand smoothing reduces entry and exit by roughly 25%, but it has little effect on the rate at which existing plants change size.
- **Main market-structure result.** Demand smoothing has a large market expansion effect. It raises the number of plants by about 39%, shifts the plant-size distribution toward larger plants, and reduces monopoly-market prevalence.
- **Welfare result.** Consumers gain from more competition, incumbent producers lose, and potential entrants lose in the model because more entry erodes rents and raises costs.
- **Economic takeaway.** Demand volatility does not simply create more turnover. In industries with large sunk and adjustment costs, volatility changes the long-run cross-sectional structure of the industry.

2 Incremental contribution of the paper

- **Demand volatility in a structural dynamic oligopoly.** Prior dynamic industry studies focused on entry, exit, productivity, and competition. This paper places local demand volatility at the center of the dynamic investment problem and quantifies its industry-structure consequences.
- **Local-market structural model with plant-size investment.** The model does not only ask whether plants enter or exit. It allows plants to choose size categories and to pay transition costs when expanding or shrinking, which is essential for understanding industry composition.
- **Plant-level Census panel in a sharply segmented market.** Ready-mix concrete provides a rare combination of homogeneous product, local geographic markets, and rich plant histories. This makes the empirical setting unusually well suited for dynamic oligopoly estimation.
- **Indirect inference CCP implementation.** The paper uses a tractable estimation strategy that combines conditional choice probabilities with simulation-based indirect inference to handle a rich dynamic game.
- **Demand-smoothing counterfactual with general-equilibrium market-structure effects.** The main policy experiment does not merely change short-run volatility. It changes firms' expectations, market profitability, entry incentives, market concentration, and plant-size composition.
- **Separates turnover from industry composition.** A key insight is that volatility can have modest effects on annual turnover but large effects on the cross-section of markets and plant sizes.

- **Fiscal-policy relevance.** The counterfactual resembles sequencing government construction contracts. The paper therefore links industrial organization to public finance and countercyclical spending design.
- **New welfare accounting for demand smoothing.** The paper decomposes the effects of smoothing into consumer surplus, incumbent producer surplus, and potential entrant surplus, highlighting distributional consequences.

3 Data and measurement

3.1 Product and local-market definition

- Ready-mix concrete must be delivered quickly after mixing; this creates highly local markets.
- The paper treats counties as independent markets because ready-mix trucks generally travel short distances and concrete is cheap relative to weight.
- Market concentration is high: most counties have fewer than six active plants and many have zero or one plant.
- Competition matters mainly through the first few competitors, which motivates a model with discrete local rivalry effects.

Interpretation: The industry is almost ideal for local oligopoly analysis. Since transportation is costly and the product is homogeneous, market structure can be tied to local demand rather than to national product differentiation.

3.2 Demand measure

- Demand is proxied by county construction employment.
- Construction demand is treated as exogenous to ready-mix market structure because concrete is a small input share in construction and concrete prices are unlikely to determine county construction employment.
- Government infrastructure spending is a major source of demand volatility.
- Demand is persistent but volatile: short-run fluctuations are large and long-run movements remain important.

Interpretation: The demand process has both persistent and transitory components. This is why smoothing short-run volatility while preserving long-run changes is meaningful.

3.3 Plant states and transitions

- Plant state includes current plant size and past plant size.
- Size categories are out, small, medium, and large.
- Past size matters because plant-size transition costs depend on history; a plant that used to be large may face a different cost of becoming large again.
- Table II documents that plants are persistent: small plants mostly stay small and large plants mostly stay large.

Interpretation: The model needs history dependence because discrete investment is not memoryless. Expansion and contraction costs depend on prior size.

3.4 Census data

- The paper uses U.S. Census plant data from the Longitudinal Business Database and Census of Manufactures data.
- The core estimation sample covers ready-mix plants from 1976 to 1999.
- County-year outcomes include the number of plants, plant employment, payroll, shipments, value added, and construction-sector employment.
- The model restricts attention to counties with manageable state spaces by excluding very large counties.

Interpretation: The detailed plant histories allow the paper to estimate entry, exit, and plant-size transition behavior, not just static concentration.

4 Models and empirical specifications

4.1 Dynamic oligopoly state

Each market has a state vector

$$s^t = \{s_1^t, s_2^t, \dots, s_N^t, M^t\},$$

where each firm-specific state contains plant-size information and idiosyncratic private shocks, and M^t is market-level demand.

Interpretation: The state captures both market demand and the composition of incumbent and potential plants. This is the object that firms use to choose entry, exit, and size.

4.2 Firm actions

- A potential entrant chooses whether to enter and at what size.
- An incumbent chooses whether to exit, remain, expand, or contract.
- Actions are discrete, matching the observed size categories.
- Firms choose policies as Markov-perfect strategies conditional on the state.

Interpretation: Discrete investment makes the model realistic for ready-mix plants, where moving from small to medium or large requires large discrete capital expenditures.

4.3 Per-period profits

A reduced-form profit function depends on plant size, demand, and competition:

$$\pi_i(s^t) = \pi(\text{size}_i^t, M^t, \text{competitors}^t; \theta) + \varepsilon_i^t.$$

- Demand raises profits, especially for larger plants.
- Competitors reduce profits, with the first competitor having a particularly large effect.
- Fixed costs differ by size.

Interpretation: The profit function is designed to capture local oligopoly effects while remaining tractable enough for a dynamic game.

4.4 Investment and transition costs

- Entry into small, medium, or large size requires sunk costs.
- Expanding or shrinking size requires transition costs.
- Costs depend on current and previous size, allowing partial reversibility.
- Large transition costs explain why firms do not react aggressively to short-run demand shocks.

Interpretation: This is the structural mechanism behind the modest turnover effect of demand smoothing. If adjustment is expensive, firms wait rather than chase temporary demand movements.

4.5 Estimation

- A first-stage model estimates demand transitions and conditional choice probabilities.
- An indirect inference step chooses structural parameters to match empirical choice probabilities and industry moments.
- The model is evaluated by simulating market histories from observed initial conditions.
- Table VII reports model fit by comparing real data and simulated data moments.

Interpretation: The estimation strategy is built to make a rich dynamic game computationally feasible while preserving the policy counterfactual structure.

5 Identification and empirical design

5.1 Core identifying variation

- The key exogenous driver is county-level construction employment.
- Ready-mix demand is local and plausibly too small to affect overall construction employment.
- Cross-county and within-county variation in demand identifies how entry, exit, and plant-size transitions respond to market size.
- Plant transition histories identify sunk and adjustment costs.

Interpretation: The empirical design uses local demand shocks as variation in market size and combines that with plant histories to infer dynamic costs.

5.2 Unobserved market heterogeneity

- Counties may differ persistently in unobserved attractiveness.
- Table IV tests for market unobservables using binary logit specifications with market-fixed effects and market-category effects.
- Market-category effects are used in subsequent estimation to handle persistent unobserved differences.

Interpretation: Without controlling for market heterogeneity, high-demand markets could be confused with persistently high-profit markets.

5.3 Why the counterfactual is informative

- The demand-smoothing counterfactual changes only the time profile of demand, not long-run average demand.
- This isolates the role of short-run volatility from secular market growth or decline.
- Comparing unsmoothed, five-year-smoothed, constant-demand, and myopic-belief cases separates volatility, long-run dispersion, and expectations.

Interpretation: The counterfactual is not a simple comparative static. It changes both realized demand volatility and firms' beliefs about future demand.

5.4 Limitations

- Demand is proxied by construction employment, not concrete orders directly.
- County markets are an approximation; some border effects may remain.
- The state space requires dropping very large markets.
- The welfare calculations rely on model assumptions about potential entrants and producer surplus.

Interpretation: The paper is strongest as a structural counterfactual study of a well-chosen local industry, not as a reduced-form causal estimate of a single shock.

6 Section-by-section study guide

- **Introduction.** Read for the main tension: demand smoothing lowers turnover, but increases the number and size of plants.
- **Industry and data.** Focus on why ready-mix concrete is local, oligopolistic, and exposed to construction demand.
- **Model.** Track the state variables: plant size, past size, idiosyncratic shocks, market demand, and competition.
- **Estimation.** Understand why CCP and indirect inference are used to make the dynamic game feasible.
- **Results.** Use Tables IV-VII and Figure 3 to see how demand, competition, and adjustment costs shape market structure.
- **Counterfactuals.** Tables VIII-X contain the main policy lessons: smoothing changes composition and surplus more than annual turnover.

7 Tables and figures

7.1 Figure 1 and Figure 2: Local oligopoly and price competition

Read:

- Figure 1 shows the spatial dispersion of ready-mix concrete plants in the Midwest.
- Figure 2 shows that median prices fall sharply with the first few competitors, then flatten.
- Together, these figures motivate local county markets and oligopoly competition.

Detailed interpretation:

- Figure 1 shows that the industry is spatially fragmented because transportation constraints prevent a national market.
- Figure 2 shows that the number of competitors matters most at low counts. This is consistent with the paper's emphasis on small local oligopolies.
- The flattening after three or four competitors motivates a profit function in which early competitors have especially large effects.

Economic message: Ready-mix competition is local and nonlinear: moving from monopoly to duopoly matters much more than adding another competitor to an already competitive market.

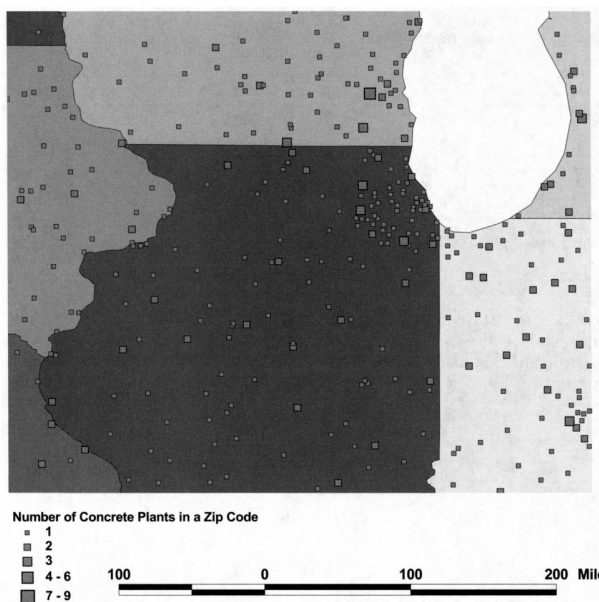


FIGURE 1.—Dispersion of ready-mix concrete plant locations in the Midwest. Source: 2 Business Patterns publicly available data set at http://www.census.gov/epcd/www/zbp_base.htm for NAICS Code 327300.

(a) Figure 1: Dispersion of ready-mix concrete plant locations.

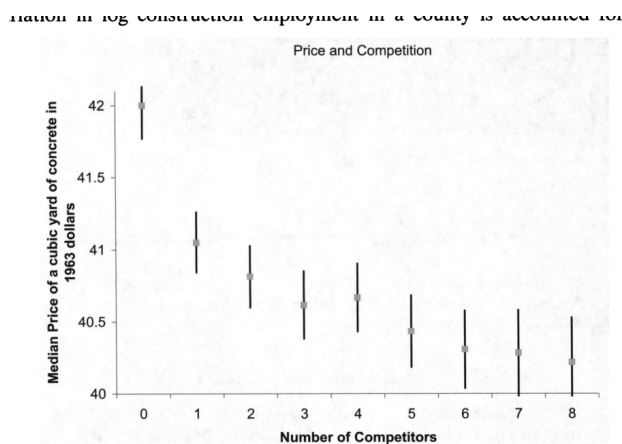


FIGURE 2.—Price declines with the addition of the first competitors, but drops by very little thereafter. Bars represent 95% confidence interval on median price.

(b) Figure 2: Price and competition.

7.2 Table I and Table II: Market concentration and plant transition dynamics

Read:

- Table I shows most county-years have fewer than six ready-mix plants; 30% have zero and 31% have one.

- Table II shows strong persistence in plant size: small plants stay small and large plants stay large with high probabilities.
- Exit from small plants is much more common than exit from large plants.

Detailed interpretation:

- Table I establishes that the empirical environment is dominated by sparse local markets, not thick competitive markets.
- Table II demonstrates why a dynamic model is needed. Static entry regressions cannot capture persistence, ramping up, ramping down, or re-entry after past size changes.
- The “past medium” and “past large” rows make adjustment-history dependence visible.

Economic message: The industry is both concentrated and sticky. Firms do not rapidly reset their plant size in response to every demand shock.

TABLE I
MOST COUNTIES IN THE UNITED STATES ARE SERVED BY FEWER THAN SIX READY-MIX CONCRETE PLANTS

Number of Concrete Plants	Number of Counties/Years	Percent
0	22,502	30%
1	23,276	31%
2	12,688	17%
3	6373	9%
4	3256	4%
5	1966	3%
6	1172	2%
More than 6	3205	4%
Total	74,438	

(a) Table I: Counties by number of concrete plants.

TABLE II
AVERAGE YEARLY PLANT TRANSITION PROBABILITIES

Current Size	Size Next Year				Total
	Out	Small ^a	Medium ^b	Large ^c	
Out	98.5%	1.2%	0.2%	0.1%	15,134
Small	8.0%	82.4%	8.1%	1.5%	1135
Small, Medium in Past	7.9%	73.6%	17.6%	1.0%	417
Small, Large in Past	11.7%	65.8%	16.4%	6.1%	140
Medium	3.2%	20.1%	68.6%	8.1%	686
Medium, Large in Past	3.2%	11.0%	64.4%	21.3%	307
Large	2.7%	4.1%	11.1%	82.1%	913

^aSmall: Less than 8 Employees.

^bMedium: 8 to 17 Employees.

^cLarge: More than 17 Employees.

(b) Table II: Plant transition probabilities.

7.3 Table III and Table IV: Summary data and market unobservables

Read:

- Table III gives county-aggregated data, with an average of 1.86 plants per market and large variation in construction employment.
- Table IV estimates plant activity logit models with market-fixed and market-category effects.
- Demand raises activity, while competitors reduce activity; the first competitor has the largest negative effect.

Detailed interpretation:

- Table III shows the breadth of demand variation. Construction employment ranges from very small counties to counties with thousands of workers.
- Table IV is a diagnostic for unobserved market heterogeneity. Adding market-category effects changes estimates and motivates using market categories in later modeling.
- The strong negative first-competitor coefficient foreshadows the structural estimates in Table VI.

Economic message: Market size supports activity, but competition erodes profits. The empirical model must separate high-demand markets from persistently attractive markets.

	Observations	Mean	Standard Deviation	5th Percentile	95th Percentile
<i>Concrete Plant Data</i>					
Concrete Plants	74,435	1.86	3.24	0	6
Employment	74,435	27.24	79.03	0	110
Payroll (in 000's)	74,435	4238	74,396	0	3600
Total Value of Shipment (in 000's)	24,677	3181	12,010	0	14,000
Value Added (in 000's)	24,677	1408	5289	0	6500
Total Assets Ending (in 000's)	24,677	1090	14,134	0	4700
<i>Construction Establishment Data</i>					
Employment	69,911	1495	5390	11	6800
Payroll (in 000's)	69,911	37,135	163,546	110	160,000
County Area (in square miles)	72,269	1147	3891	210	3200

(a) Table III: County-aggregated summary statistics.

Dependent Variable: Activity	I	II (FE [†])	III	IV (μ)	V	VI	VII	VIII
Log County Construction	0.17*** (0.01)	0.03 (0.02)	0.19*** (0.01)	0.02* (0.01)	0.03*** (0.01)	0.20*** (0.01)	0.09*** (0.02)	0.07*** (0.01)
Employment	-1.40*** (0.05)	-1.26*** (0.04)	-0.85*** (0.04)	-1.07*** (0.04)	-1.01*** (0.04)	-0.69*** (0.01)	-0.73*** (0.04)	-0.85*** (0.04)
First Competitor	0.00 (0.04)	-0.54*** (0.04)	-0.03 (0.04)	-0.48*** (0.04)	-0.47*** (0.04)	-0.03 (0.04)	0.04 (0.04)	-0.03 (0.04)
Second Competitor	0.03 (0.04)	-0.33*** (0.04)	0.04 (0.05)	-0.32*** (0.05)	-0.32*** (0.05)	0.00 (0.06)	0.07 (0.05)	0.03 (0.05)
Third Competitor	0.02 (0.03)	-0.13*** (0.03)	0.09** (0.03)	-0.06* (0.03)	-0.10* (0.03)	0.10** (0.05)	0.10** (0.03)	0.09** (0.03)
Log Competitors Above 4	6.89*** (0.04)	6.50*** (0.03)	6.92*** (0.04)	6.73*** (0.04)	6.75*** (0.04)	7.06*** (0.05)	6.89*** (0.04)	6.91*** (0.04)
Small	6.85*** (0.05)	6.36*** (0.04)	6.82*** (0.05)	6.60*** (0.05)	6.61*** (0.05)	6.92*** (0.06)	6.85*** (0.05)	6.83*** (0.05)
Small, Medium in Past	6.41*** (0.07)	5.90*** (0.06)	6.34*** (0.07)	6.19*** (0.07)	6.19*** (0.07)	6.42*** (0.08)	6.40*** (0.07)	6.36*** (0.07)
Small, Large in Past	7.72*** (0.06)	7.34*** (0.05)	7.66*** (0.06)	7.54*** (0.06)	7.56*** (0.06)	7.95*** (0.08)	7.72*** (0.06)	7.66*** (0.06)
Medium	7.72*** (0.08)	7.26*** (0.08)	7.60*** (0.08)	7.47*** (0.08)	7.47*** (0.08)	7.82*** (0.10)	7.71*** (0.08)	7.63*** (0.05)
Medium, Large in Past	7.85*** (0.06)	7.48*** (0.05)	7.71*** (0.06)	7.63*** (0.06)	7.64*** (0.06)	7.91*** (0.07)	7.83*** (0.06)	7.73*** (0.06)
Large								
Market-Fixed Effects		X						
State-Fixed Effects			X					
Year-Fixed Effects			X					
Market Classification Variable ^o								
Average Number of Plants				X				
Lagged Average Plants					X			
Before 1983 Average Plants						X		
Construction Employment							X	
Total Shipments of Concrete								X
Observations	409,850	409,850	409,850	260,170	409,850	409,850	409,850	409,850
Markets	2029	2029	2029	2029	2029	2029	2029	2029
Log-Likelihood	-37,541	-32,759	-37,429	-36,713	-36,695	-22,230	-37,524	-37,384
χ^2	49,545	301,019	51,517	52,775	51,795	34,019	50,302	51,538

^aStandard errors are clustered by market. [†] Market-fixed effects are implemented via a conditional logit. ^o Average number of plants (μ) is the mean number of plants in a market, rounded to the nearest integer. Lagged Average Plants is the mean number of plants in the market for years preceding t , rounded to the nearest integer. Before 1983 Average Plants is the mean number of plants in a market before 1983, rounded to the nearest integer. Only years from 1983 to 1999 are used in the regression for this market classification variable in Column V. Construction Employment Classification and Total Shipments of Concrete use the mean of these variables in a market to classify markets into four categories. *, **, *** indicate statistical significance at the 5%, 1%, and 0.1% levels, respectively.

(b) Table IV: Active-plant logit regressions.

7.4 Table V and Table VI: Plant-size choices and structural estimates

Read:

- Table V estimates multinomial logits for choosing small, medium, or large status next year.
- Table VI reports structural estimates of fixed costs, demand effects, competition effects, transition costs, and shock variance.
- Fixed costs rise with plant size, demand effects rise with plant size, and the first competitor has a large negative profit effect.
- Transition costs are large, especially for entry and large size changes.

Detailed interpretation:

- Table V is the reduced-form bridge to the structural model. It shows which state variables predict future size choices.
- Table VI gives the economic primitives. Small plants have lower fixed costs but smaller demand sensitivity; large plants have higher fixed costs and stronger demand sensitivity.
- The transition-cost estimates explain why demand fluctuations do not mechanically translate into high annual turnover.

Economic message: The model's core tradeoff is visible here: larger plants benefit more from demand, but they are costly to build and costly to adjust.

TABLE V
MULTINOMIAL LOGIT ON THE CHOICE TO BE LARGE, MEDIUM, OR SMALL

Dependent Variable	Independent Variable	I		II (Market Category)	
		Coeff.	S.E.	Coeff.	S.E.
<i>Small</i> in $t + 1$	Small	6.59	(0.03)	6.42	(0.03)
	Small, Medium in Past	6.45	(0.04)	6.18	(0.04)
	Small, Large in Past	5.94	(0.06)	5.72	(0.06)
	Medium	6.02	(0.05)	5.81	(0.05)
	Medium, Large in Past	5.41	(0.08)	5.16	(0.08)
	Large	4.58	(0.06)	4.37	(0.06)
	Log County Employment	0.13	(0.01)	-0.06	(0.01)
	First Competitor	-1.42	(0.04)	-1.71	(0.04)
	Second Competitor	0.10	(0.03)	-0.46	(0.03)
	Third Competitor	0.16	(0.04)	-0.26	(0.04)
	Log of Competitors Above 3	0.11	(0.03)	-0.04	(0.03)
	Market Category μ			X	
	Constant	-3.94	(0.06)	-3.17	(0.06)
<i>Medium</i> in $t + 1$	Small	6.25	(0.05)	6.08	(0.05)
	Small, Medium in Past	6.96	(0.06)	6.70	(0.06)
	Small, Large in Past	6.43	(0.08)	6.22	(0.08)
	Medium	9.16	(0.06)	8.96	(0.06)
	Medium, Large in Past	9.08	(0.08)	8.83	(0.08)
	Large	7.44	(0.07)	7.23	(0.07)
	Log County Employment	0.29	(0.01)	0.12	(0.01)
	First Competitor	-1.54	(0.05)	-1.87	(0.05)
	Second Competitor	0.00	(0.04)	-0.53	(0.04)
	Third Competitor	0.06	(0.05)	-0.32	(0.05)
	Log of Competitors Above 3	0.02	(0.03)	-0.11	(0.03)
	Market Category μ			X	
	Constant	-6.72	(0.08)	-5.99	(0.09)
<i>Large</i> in $t + 1$	Small	5.04	(0.08)	4.88	(0.08)
	Small, Medium in Past	4.53	(0.13)	4.28	(0.13)
	Small, Large in Past	5.78	(0.11)	5.58	(0.11)
	Medium	7.46	(0.08)	7.27	(0.08)
	Medium, Large in Past	8.37	(0.09)	8.13	(0.09)
	Large	9.76	(0.07)	9.56	(0.07)
	Log County Employment	0.52	(0.01)	0.34	(0.02)
	First Competitor	-1.61	(0.05)	-1.94	(0.06)
	Second Competitor	-0.03	(0.05)	-0.58	(0.05)
	Third Competitor	-0.02	(0.06)	-0.42	(0.06)
	Log of Competitors Above 3	-0.04	(0.04)	-0.17	(0.04)
	Market Category μ			X	
	Constant	-8.58	(0.11)	-7.83	(0.11)
	Observations	409,850		409,850	
	Log-Likelihood	84,855		83,814	
	Likelihood Ratio	400,760		402,841	

(a) Table V: Multinomial logit size choice.

TABLE VI
ESTIMATES FOR THE DYNAMIC MODEL OF ENTRY, EXIT, AND INVESTMENT^a

		Coeff.	S.E.*
Fixed Cost	Small	-139	(6)
	Medium	-244	(10)
	Large	-285	(6)
Log Construction Employment	Small	20	(1)
	Medium	35	(2)
	Large	45	(1)
1st Competitor	Small	-48	(4)
	Medium	-58	(5)
	Large	-63	(6)
Log Competitors (Above 1)	Small	-17	(3)
	Medium	-44	(4)
	Large	-48	(3)
<i>Transition Costs</i>			
Out $\rightarrow$ Small		-1002	(11)
	Out $\rightarrow$ Medium [†]	-2000	(107)
	Out $\rightarrow$ Large	-1771	(53)
Small $\rightarrow$ Medium		-332	(7)
	Small, Past Medium $\rightarrow$ Medium	-772	(32)
	Small, Past Large $\rightarrow$ Medium	-325	(8)
Small $\rightarrow$ Large		-1809	(73)
	Small, Past Medium $\rightarrow$ Large	-608	(19)
	Small, Past Large $\rightarrow$ Large	-343	(16)
Medium $\rightarrow$ Small		-107	(6)
	Medium, Past Large $\rightarrow$ Small	-314	(6)
	Medium $\rightarrow$ Large	101	(14)
Medium, Past Large $\rightarrow$ Large		-43	(7)
	Large $\rightarrow$ Small	-254	(7)
	Large $\rightarrow$ Medium	-403	(6)
Standard Deviation of Shock		133	

^aAll estimates in thousands of dollars. [†] The entry costs of a medium-sized plant are calibrated to \$ 2 million. * Standard Errors are computed using 100 block bootstrap replications, where I reestimate the demand transition process $\hat{D}^{\mu}$ and the conditional choice probabilities $\hat{P}^{\mu}$, then minimize the criterion function Q to find $\hat{\theta}$. I block bootstrap by market, resampling a market's history from 1976 to 1999, so the computed standard errors account for serial correlation within a market.

(b) Table VI: Dynamic model estimates.

7.5 Figure 3 and Table VII: Market size, plant size, and model fit

Read:

- Figure 3 shows that larger markets have both larger plants and more plants.
- The number of plants is a concave function of log construction employment.
- Table VII compares real data to simulated model moments.
- The model matches size distribution and market structure reasonably well, but underpredicts entry and exit relative to the data.

Detailed interpretation:

- Figure 3 is crucial for counterfactuals: because market structure responds concavely to demand, reducing intertemporal volatility can affect expected profitability and market size.
- Table VII validates the model before policy experiments. The model captures the number of plants, size shares, and market structure better than it captures all turnover.
- The gap in entry/exit suggests idiosyncratic shocks and plant-specific volatility matter beyond market demand alone.

Economic message: The model is good enough to study long-run market-structure effects, but the underprediction of turnover cautions against overstating demand's role in all entry and exit.

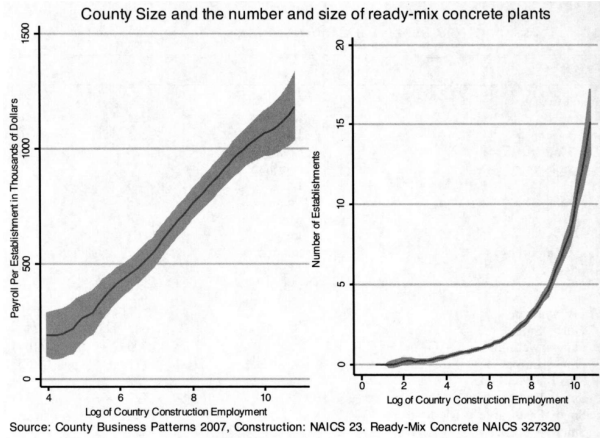


FIGURE 3.—Larger markets have both more plants and larger plants. Local polynomial regressions with shaded areas representing the 95% confidence interval.

(a) Figure 3: County size, number of plants, and plant size.

TABLE VII
MODEL FIT

	I		II		III	
Moments	Real Data (1976–1999)		Simulated Data Using Model $\hat{\theta}^a$		Simulated Data Using CCP $\hat{p}^{\mu a}$	
<i>Plant-Level Moments</i>						
Share of Small Plants	48%	(1%)	53%	(1%)	52%	(1%)
Share of Medium Plants	27%	(0%)	23%	(1%)	26%	(1%)
Share of Large Plants	25%	(1%)	24%	(1%)	22%	(1%)
Entry Rate	5.8%	(0.0%)	2.9%	(0.2%)	7.0%	(0.2%)
Exit Rate	5.4%	(0.0%)	2.9%	(0.2%)	7.3%	(0.2%)
Ramping Up Rate	10%	(0.1%)	10%	(0.3%)	11%	(0.2%)
Ramping Down Rate	9%	(0.1%)	10%	(0.5%)	9%	(0.2%)
<i>Market-Level Moments</i>						
Number of Plants per Market	2.0	(0.2)	2.0	(0.4)	2.3	(0.1)
No Plants in Market	2%	(0%)	4%	(1%)	18%	(1%)
Monopoly Market	46%	(1%)	43%	(1%)	32%	(0%)
Duopoly	26%	(1%)	29%	(1%)	18%	(0%)
More Than 2 Plants	26%	(1%)	24%	(1%)	32%	(1%)
Number of Plants in Category 1	1.08	(0.00)	1.18	(0.02)	1.42	(0.06)
Number of Plants in Category 2	1.76	(0.01)	1.80	(0.05)	2.20	(0.06)
Number of Plants in Category 3	2.62	(0.01)	2.32	(0.08)	2.98	(0.08)
Number of Plants in Category 4	4.15	(0.04)	4.25	(0.16)	4.44	(0.10)
Coefficient of Variation						
Number of Plants Within Market	0.7	(0.1)	0.6	(0.2)	1.7	(0.0)
Correlation Demand and Plant Size	0.23	(0.01)	0.26	(0.01)	0.23	(0.01)
Correlation Demand and Number of Plants	0.54	(0.01)	0.67	(0.01)	0.28	(0.01)

^aData are simulated using either II—computed policies given estimated parameters $\hat{\theta}$ or III—estimated conditional choice probabilities $\hat{p}^{\mu}$, taking as an initial condition the markets in 1976. The market block bootstrapped

(b) Table VII: Model fit.

7.6 Table VIII and Table IX: Demand smoothing and industry composition

Read:

- Table VIII compares unsmoothed demand, five-year smoothing, constant demand, and myopic beliefs.
- Five-year smoothing reduces entry and exit but raises total plants from 3,643 to 5,433.
- Table IX shows that five-year smoothing lowers monopoly markets from 43% to 25% and raises markets with more than two plants from 25% to 46%.
- Five-year smoothing also shifts the industry toward medium and large plants.

Detailed interpretation:

- Table VIII shows the subtle dynamic result: smoothing lowers turnover rates, but because it raises the number of plants, total investment rises.
- Table IX shows the market expansion effect: smoothing makes more markets viable and more competitive.
- The difference between constant demand and five-year smoothing matters because constant demand also changes cross-sectional demand dispersion, not just volatility over time.

Economic message: Demand smoothing changes the industry's cross-section more than its year-to-year adjustment rate. It creates more plants and more competitive local markets.

	Unsmoothed Demand ($\bar{D}^u$)	5 Years of Smoothing	Constant Demand	Firms Believe Demand is Constant
<i>Turnover</i>				
Entry Rate	2.7%	2.2%	2.2%	4.1%
Exit Rate	2.9%	2.0%	2.1%	4.5%
Change in Size Rate	20%	18%	17%	18%
<i>Investment</i>				
Sunk Entry Costs per Year (in Million \$)	132	137	107	155
Size Changing Costs per Year (in Million \$)	307	496	407	337
Total Plants	3643	5433	4264	3879

(a) Table VIII: Demand smoothing, turnover, and size changing.

	Unsmoothed Demand	Constant Demand	5 Years of Smoothing
Total Plants	3645	4264	5433
Fixed Costs (per Period in Millions of \$)	717	878	1109
<i>Industry Composition</i>			
Small Plants	54%	48%	49%
Medium Plants	23%	23%	24%
Big Plants	23%	29%	28%
<i>Market Structure</i>			
Markets With no Plants	5%	8%	1%
Markets With 1 Plant	43%	36%	25%
Market With 2 Plants	28%	24%	29%
Markets With More Than 2 Plants	25%	32%	46%

(b) Table IX: Demand smoothing and industry composition.

7.7 Table X: Welfare Effects of Demand-Smoothing Policies

Read:

- Table X reports net present value changes under five years of demand smoothing.
- Consumer surplus increases by \$860 million.
- Incumbent producer surplus falls by \$609 million.
- Potential entrant surplus falls by \$36 billion in the model.

Detailed interpretation:

- The consumer gain comes from more competition and lower prices in markets that become less concentrated.
- Incumbents lose because additional entry erodes market power.
- The large potential-entrant number reflects how the model treats many potential entrants who usually do not enter; it should be interpreted cautiously.

Economic message: Demand smoothing has a distributional dimension: consumers gain from competition, while producers and potential entrants bear costs through lower rents and more entry pressure.

TABLE X
WELFARE EFFECTS OF DEMAND-SMOOTHING POLICIES^a

Change in Net Present Value of	
Consumer Surplus	\$860 Million
Producer Surplus for Incumbents	–\$609 Million
Producer Surplus for Potential Entrants	–\$36 Billion

^aNumbers in this table refer to the difference in the net present value of surplus (using a 5% discount rate) between five years of smoothing and unsmoothed demand, averaged between 25 and 50 years after the policies were put into place, using 1976 as an initial state.

Table X: Welfare Effects of Demand-Smoothing Policies.

8 Result map

- **Industry structure.** Ready-mix concrete markets are local oligopolies because the product is heavy, cheap, and perishable.
- **Reduced form.** Demand raises activity and plant size, while competitors reduce profits.
- **Structural costs.** Entry and size-change costs are large, especially for entering or moving to a large plant.
- **Model fit.** The model fits market structure and size distribution but underpredicts some entry and exit.
- **Counterfactual.** Five-year demand smoothing reduces turnover but raises the number of plants sharply.
- **Composition.** Smoothing creates more multi-plant markets and shifts plant sizes upward.
- **Welfare.** Consumers gain from more competition; incumbent producers lose; potential entrant surplus is strongly negative in the model accounting.

9 Short summary

- Collard-Wexler studies how demand volatility affects the ready-mix concrete industry.
- The industry is well suited for the analysis because markets are local and oligopolistic.
- Demand is proxied by county construction employment.
- The paper estimates a dynamic model of entry, exit, and discrete plant-size investment using Census plant histories.
- Demand smoothing reduces entry and exit by about 25% but does not change the rate at which plants change size.
- Smoothing has a large market expansion effect: it raises the number of plants, lowers monopoly prevalence, and shifts the industry toward larger plants.
- Consumer surplus rises, incumbent producer surplus falls, and potential entrant surplus falls substantially under the model's accounting.
- The paper's key lesson is that volatility can reshape the industry cross-section even when short-run turnover effects are limited.

10 Conclusion

10.1 Core conclusion

Collard-Wexler shows that demand fluctuations matter for industry structure in ways that are not captured by simple entry and exit rates. In the ready-mix concrete industry, short-run demand volatility interacts with sunk entry costs, size-change costs, and local oligopoly competition to determine the long-run number and size of plants.

10.2 Economic intuition

When demand is highly volatile, firms discount short-run movements because they are less informative about future profitability. Large sunk and adjustment costs make it unattractive to chase temporary demand shocks. When demand is smoothed, remaining demand changes become more informative, and lower volatility changes the profitability of markets through concavity in the relationship between demand and market structure.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to connect demand volatility, dynamic oligopoly, and policy counterfactuals in a structural empirical model. It shows that the consequences of volatility can appear in the cross-section of market structure and plant sizes, not only in more volatile investment.

10.4 Policy implications

The demand-smoothing counterfactual resembles countercyclical fiscal planning. Sequencing government construction contracts can reduce turnover and increase competition, but it can also increase total investment costs and redistribute surplus from producers to consumers.

10.5 Limitations

The county market definition is an approximation, demand is measured with construction employment, and the model requires a finite state space. The welfare calculation for potential entrants is especially model-dependent. Still, the paper provides a rare structural counterfactual for volatility in a concentrated local industry.

10.6 Takeaway

Demand volatility is not merely noise around a trend. In industries with costly entry and adjustment, it shapes market structure, plant size, producer rents, and consumer surplus.